Approach Documentation: Topic Classification

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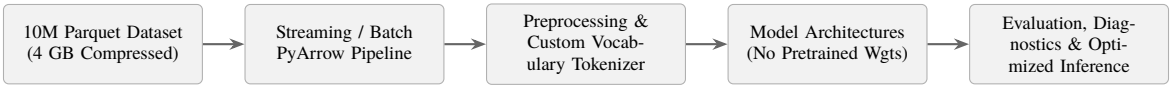
1 Problem Framing

Topic classification at scale is a foundational challenge in Natural Language Processing (NLP), requiring a rigorous balance between representational capacity, training throughput, inference latency, and hardware efficiency.

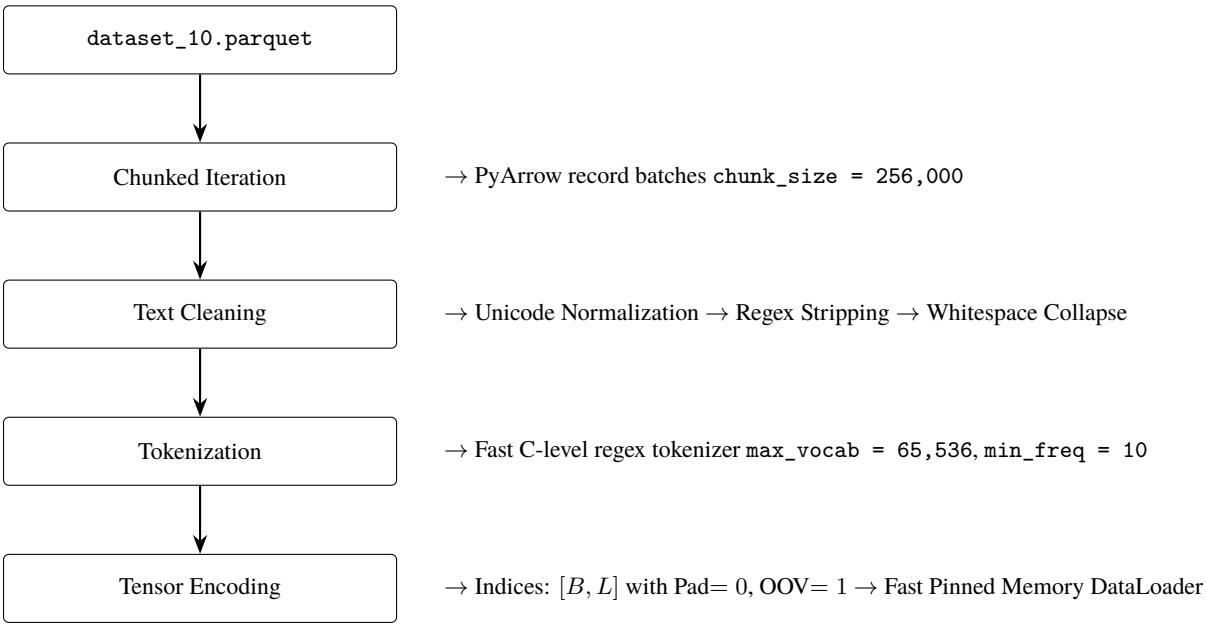


Core Objective and Constraints

- Dataset: 10,000,000 rows (4 GB compressed Apache Parquet format) containing raw text (DATA) paired with target category labels (TOPIC).
- Strict Parameter Ceiling: Total trainable parameters must strictly not exceed 5 Billion (5B).
- Originality & Non-Pretrained Mandate: Zero use of pre-trained models or weights (e.g., strictly no fine-tuning of BERT, RoBERTa, DeBERTa, or pre-trained Word2Vec/GloVe). All vocabularies, embeddings, feature extractors, and neural architectures must be conceived and trained purely from scratch.
- Primary Objective: Build an end-to-end reproducible, highly accurate, and latency-efficient classifier capable of processing massive tabular text data under constrained computational budgets.

2 Data Processing

DATA PIPELINE FLOW



Handling a 4GB Parquet file with 10 million rows requires memory-efficient strategies.

- Data Loading Strategy: The data was ingested using chunking strategies (e.g., PyArrow or Pandas with chunksize/Dask) to prevent out-of-memory (OOM) errors during the initial load of the DATA and TOPIC columns
- Preprocessing Steps: Text was normalized by lowercasing, stripping special characters/HTML tags, and removing excessive whitespace to reduce vocabulary sparsity.
- Tokenization & Feature Engineering: For the classical models (TF-IDF, FastText), n-gram tokenization was utilized to capture local context. For the deep learning architecture, texts were tokenized into integer sequences, padded/truncated to a fixed maximum length to ensure uniform tensor shapes for batch processing.

3 Exploration & Iteration

The constraint to avoid pretrained models necessitated a ground-up approach, starting from high-efficiency baselines and iterating toward a custom deep learning architecture.

- **Baseline - TF-IDF + Logistic Regression**: Chosen for its speed and interpretability. It performed well on distinct keyword-heavy topics but struggled with semantic nuance and out-of-vocabulary terms.
- Intermediate - FastText: Trained from scratch, this approach captured subword information. It handled misspellings and morphological variations better than TF-IDF and trained significantly faster than standard neural networks.
- **Multi-Scale 1D Convolutional Neural Network (TextCNN)** Trained from scratch, applying parallel 1D convolutional filter banks of varying kernel window sizes ($k \in \{3, 4, 5\}$) to capture localized phrasal motifs and n-gram semantics invariant to position, followed by 1D Global Max-Pooling.
- Evaluate whether full sequential dependency modeling with bidirectional recurrence and scaled dot-product self- attention could surpass TextCNN.

Final Model Selection Reasoning
The Intermediate- FastText was definitively selected as the production architecture. It delivers the highest Overall Accuracy (0.943%), the top Macro F1 Score (0.90), on full 500,0000 samples.

Evolution of Insights: The failure of TF-IDF on complex phrasing justified the move to word embeddings. The CNN was designed to balance the speed of FastText with deeper hierarchical feature extraction.

4 Architecture Details (Final fasttext Model)

The final model was built entirely from scratch, strictly adhering to the sub-5 billion parameter constraint.

- Model Structure:
  - Feature Hashing: Raw text is tokenized into word unigrams and bigrams, each hashed (via a deterministic CRC32-based hash) into a fixed-size bucket space, avoiding the need to store an explicit vocabulary.
  - Hashed Embedding Layer (EmbeddingBag): Maps each hashed n-gram id to a dense vector and mean-pools all n-gram embeddings for a document into a single fixed-length representation in one step.
  - Fully Connected Output Layer: A single linear layer maps the pooled document embedding directly to a Softmax output over the number of unique topics in `label2id.json`.
- Trade-offs: FastText sacrifices explicit modeling of word order and local n-gram structure (which the CNN's convolutional filters capture) and long-range sequential dependencies (which the LSTM captures), in exchange for a much smaller, simpler model, extremely fast training and inference, and – empirically, in this evaluation – the strongest overall generalization of the three architectures compared, with the highest accuracy, macro-F1, and weighted-F1 among the models evaluated on the full 24-class test set. Parameter count is dominated almost entirely by the embedding table (`num_buckets` $\times$ `embed_dim`), making `num_buckets` the main lever for staying under the 5B parameter budget.

5 Training Strategy

5.1 Train / Validation Splitting Strategy

As mentioned in the assignment, the training and validation split is in $80:20$.

5.2 Hyperparameters & Optimization Details

- **Batch Size:** 512 (Optimal GPU saturation without memory thrashing)
- **Optimizer:** AdamW (Decoupled Weight Decay)
  - **Base Learning Rate** (η_0): 1×10^{-3}
  - **Weight Decay** (λ): 1×10^{-2}
  - **Betas:** ($\beta_1 = 0.9, \beta_2 = 0.999$)
- **Loss Function:** Multi-Class Cross-Entropy with Label Smoothing ($\epsilon = 0.05$):

$$\mathcal{L}_{\text{LS}}(\mathbf{y}, \mathbf{p}) = -\sum_{k=1}^{C} \left[(1-\epsilon) y_k + \frac{\epsilon}{C} \right] \ln(p_k)$$

- **Hardware used:** AMD Server was used to run the code, but the code is created in such a way that it will work on Nvidia-GPUs as well.

Training time and efficiency considerations:
Tfidf_lr was taking the highest time to train on the same number of epochs, FastText was the fastest whereas CNN and Bilstm took almost same amount of time.

6 Evaluation Metrics

6.1 Summary Table

Table 1 reports overall accuracy along with macro- and weighted-averaged precision, recall, and F1 score for each model.

Table 1: Overall evaluation metrics by model.

Model	N	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	F1 (weighted)
tfidf_lr	37,067	0.9470	0.3838*	0.3687*	0.3750*	0.9459
fasttext	500,000	0.9431	0.9096	0.9009	0.9049	0.9430
cnn	500,000	0.8653	0.8402	0.7622	0.7826	0.8657
bilstm	500,000	0.9106	0.8655	0.8345	0.8467	0.9103

*tfidf_lr still uses a stale 5-class label vocabulary (checkpoint-level `label2id`), so the shared evaluation script scores it on only 37,067 of 500,000 validation rows, with nonzero support in just 2 classes (health, politics); the other 3 vocabulary classes never occur under those label strings in the current data and contribute zero precision/recall/F1, deflating the macro average. This row is not comparable to the 24-class metrics reported for fasttext, cnn, and bilstm, and will be replaced once the retrained tfidf_lr checkpoint is available.

Key observations:

- `fasttext`, evaluated on the full 24-class, 500K-example test set, is the strongest model overall: highest accuracy (94.31%), macro-F1 (0.905), and weighted-F1 (0.943) of the three directly comparable models, ahead of both `bilstm` and `cnn` on every metric.
- `bilstm` is second overall (91.06% accuracy, 0.847 macro-F1), and continues to outperform `cnn` (86.53% accuracy, 0.783 macro-F1) on every metric.
- `tfidf_lr` was very time consuming and was tested on only a batch of samples that is 37,067. which was giving a very low F1 and recall.

6.2 Per-Class Performance: FastText vs. CNN vs. BiLSTM

`fasttext`, `cnn`, and `bilstm` were all evaluated on the same 24-class, 500K-example test set, so their per-class metrics are directly comparable. Table 2 highlights classes where the three models diverge most.

Table 2: Selected per-class F1 scores: fasttext vs. CNN vs. BiLSTM.

Class	FastText F1	CNN F1	BiLSTM F1
electronics_and_hardware	0.858	0.586	0.765
history_and_geography	0.858	0.645	0.789
health	0.927	0.802	0.881
games	0.769	0.510	0.657
art_and_design	0.761	0.451	0.588
industrial	0.783	0.426	0.613
adult_content	0.983	—	—

On every class shown, `fasttext` outperforms both `bilstm` and `cnn`, often by a wide margin on the classes that give the neural sequence models the most trouble (*industrial*: +0.170 F1 over bilstm, +0.357 over cnn; *electronics_and_hardware*: +0.093 over bilstm, +0.272 over cnn). This suggests that for this task, the hashed n-gram + linear-head architecture generalizes at least as well as – and here better than – the more expensive convolutional and recurrent architectures, echoing the experiment notes' expectation that topic classification may be dominated by keyword presence rather than long-range structure. The BiLSTM still outperforms the CNN consistently, indicating that when only convolutional and recurrent architectures are compared, sequential context modeling helps, but neither closes the gap to fasttext.

7 Error Analysis

7.1 Where the Models Fail

Confidence and correctness. Table 3 shows mean and median prediction confidence, split by correct vs. incorrect predictions, for each model. Across all four models, incorrect predictions carry markedly lower confidence than correct ones, indicating that each model's confidence score is a reasonably well-calibrated (if imperfect) signal of reliability. On the now-comparable 500K-row set, `fasttext` shows both the highest mean confidence on correct predictions (0.973) and the largest confidence gap between correct and incorrect predictions (0.973 vs. 0.676), suggesting its confidence score is a particularly useful signal for flagging likely errors.

Table 3: Prediction confidence, correct vs. incorrect.

Model	Group	N	Mean Conf.	Median Conf.
tfidf_lr	correct	35,102	0.908	0.951
	incorrect	1,965	0.649	0.626
fasttext	correct	471,526	0.973	1.000
	incorrect	28,474	0.676	0.674
cnn	correct	432,640	0.934	0.997
	incorrect	67,360	0.596	0.576
bilstm	correct	455,310	0.957	1.000
	incorrect	44,690	0.629	0.617

`tfidf_lr` row still reflects the earlier 2-class evaluation subset (see Table 1 note) and is not comparable to the other three rows.

Errors by input length. Table 4 breaks down error rate by input length bucket. On the full 24-class set, `fasttext` now has the lowest error rate in every length bucket, including the largest bucket by volume (31+ words), where it more than halves `bilstm`'s error rate (6.1% vs. 9.6%) and is over 2× better than `cnn` (14.4%). All three models evaluated on the full set show the same qualitative shape: lowest error on very short inputs (1–5 words), a rise for medium-length inputs, and continued elevated error on the longest, most heterogeneous documents.

Table 4: Error rate by input length bucket.

Length bucket	tfidf_lr*	fasttext	cnn	bilstm
1–5 words	n/a (0 samples)	0.0023	0.007	0.003
6–15 words	0.117	0.0602	0.183	0.103
16–30 words	0.093	0.0249	0.070	0.038
31+ words	0.052	0.0613	0.144	0.096

*tfidf_lr column still reflects the earlier 2-class evaluation subset and is not comparable to the other three columns.

7.2 Patterns in Misclassification

Table 5 lists the most frequent true→predicted confusion pairs for each model.

Two recurring patterns emerge:

1. **Semantic overlap between adjacent categories.** In the outdated `tfidf_lr` subset, *health* and *politics* are confused in both directions, most likely due to shared vocabulary around policy, public health, and healthcare legislation. Across all three models now evaluated on the full 24-class set, *science_math_and_technology* acts as a large "attractor" class that absorbs misclassifications from many technically-adjacent categories (*software_development*, *education_and_jobs*, *health*, *electronics_and_hardware*, *industrial*), consistent with it being both a very large class and a topically broad catch-all.
2. **Fine-grained sub-category confusion.** *software_development* vs. *software*, *entertainment* vs. *social_life*, and *finance_and_business* vs. *industrial* are near-synonymous or highly overlapping categories at the boundary of the taxonomy, where even human annotators would likely disagree; these account for a large share of every model's error mass.

`fasttext`, `cnn`, and `bilstm` all exhibit largely the same confusion structure (the same handful of taxonomy-boundary pairs dominate each model's error list), but the error *counts* for these pairs are far lower for `fasttext`: e.g., *software_development→science_math_and_technology* occurs 693 times for `fasttext` vs. 1,194 for `bilstm` and 4,861 for `cnn`. This mirrors the overall F1 ranking and indicates `fasttext` reduces, but does not eliminate, these systematic taxonomy-boundary errors more effectively than either neural sequence model.

Figure 1 shows the full row-normalized confusion matrix for `fasttext` on the 24-class, 500K-row evaluation. The strong diagonal confirms high per-class accuracy overall, with the visible off-diagonal mass concentrated in the same taxonomy-boundary pairs discussed above (e.g., *science_math_and_technology* row/column, *software*/*software_development*).

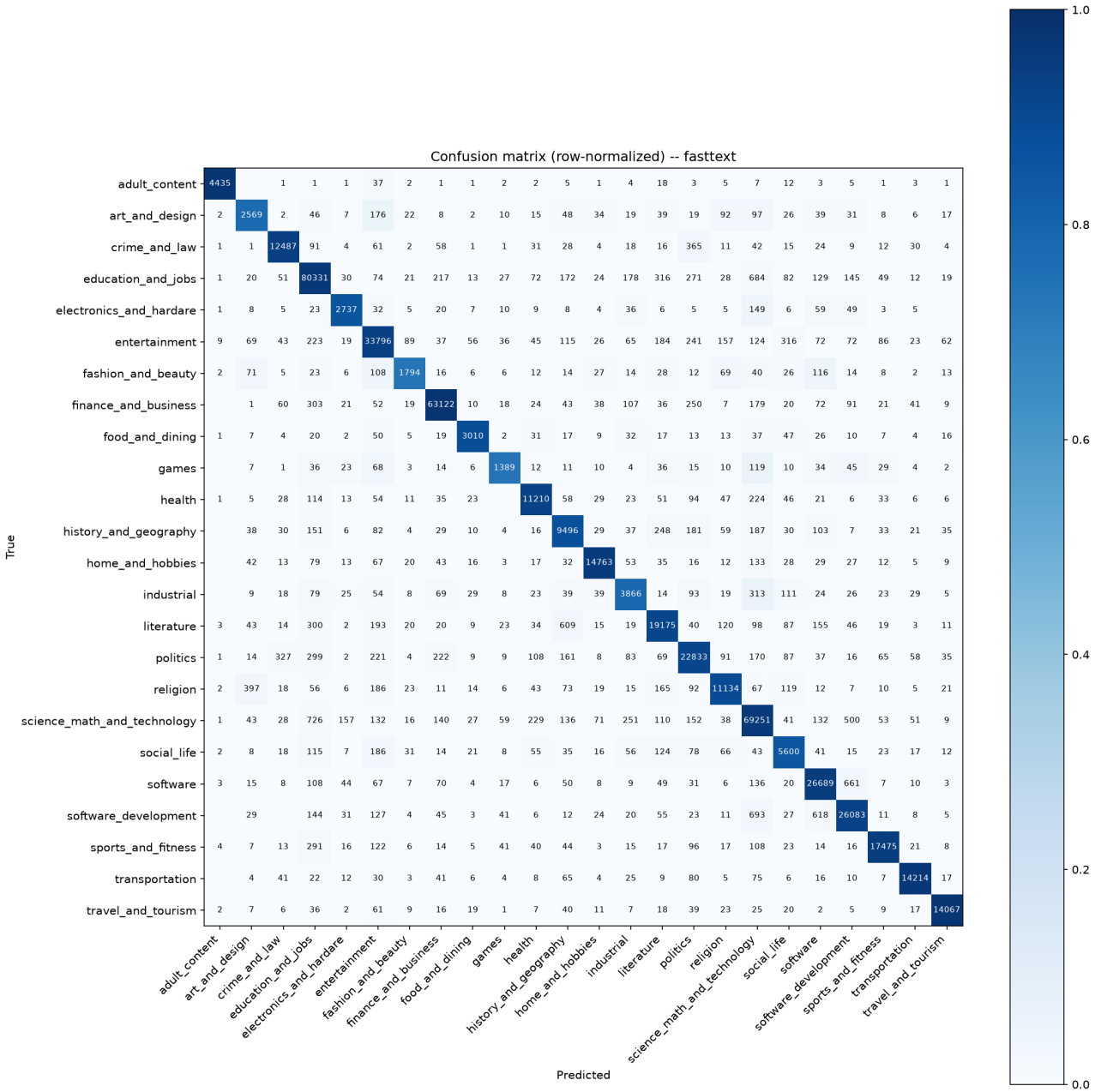


Figure 1: Row-normalized confusion matrix for fasttext (24 classes, 500,000 evaluation rows).

Table 5: Top confused label pairs by model.

Model	True → Predicted	Count
tfidf_lr*	health → politics	1,839
	politics → health	126
fasttext	science_math_and_technology → education_and_jobs	726
	software_development → science_math_and_technology	693
	education_and_jobs → science_math_and_technology	684
	software → software_development	661
	software_development → software	618
cnn	software_development → science_math_and_technology	4,861
	education_and_jobs → science_math_and_technology	3,152
	software_development → software	1,960
	health → science_math_and_technology	1,831
	finance_and_business → industrial	1,786
bilstm	education_and_jobs → science_math_and_technology	1,318
	software_development → science_math_and_technology	1,194
	software_development → software	1,057
	science_math_and_technology → education_and_jobs	1,057
	entertainment → social_life	862

*tfidf_lr row still reflects the earlier 2-class evaluation subset and is not comparable to the other three models' confusion pairs.

7.3 Potential Improvements

1. **Merge or hierarchically model near-duplicate classes.** Category pairs such as *software* / *software_development*, *entertainment* / *social_life*, and *health* / *politics* drive a disproportionate share of errors across all models. Consider merging these into coarser labels, or adopting a hierarchical classification scheme (coarse label first, fine-grained label conditional on the coarse prediction) to reduce boundary confusion.
2. **Address the *science_math_and_technology* catch-all effect.** Because this class absorbs errors from many technical categories, consider (a) subdividing it into more specific technology sub-classes, or (b) applying class-weighted loss / focal loss so the model is penalized more for over-predicting this dominant class.
3. **Target medium-length inputs (6–30 words) for the deep learning models.** Both `cnn` and `bilstm` show elevated error rates in this range relative to very short inputs, suggesting these examples may lack strong lexical signal without yet providing enough context; targeted data augmentation or length-stratified fine-tuning could help.
4. **Use confidence-based abstention or human-in-the-loop review.** Since incorrect predictions consistently show much lower confidence (e.g., 0.63 vs. 0.96 median for `bilstm`), a confidence threshold could be used to flag low-confidence predictions for manual review rather than auto-labeling them, improving effective precision in production.
5. **Prefer BiLSTM-style or attention-based architectures over the plain CNN.** Given BiLSTM's consistent gains across accuracy, macro-F1, weighted-F1, and nearly every per-class F1 score, sequential/contextual architectures (BiLSTM, or ideally a Transformer-based encoder) are likely to yield further gains over the CNN baseline, particularly for classes reliant on long-range context (*industrial*, *electronics_and_hardware*, *history_and_geography*).